Automatic Flight Control System (AFCS) (Mod 6/2303 and S.O.O. 6221) – General Data

1. General

Refer to Figure 1 for component locations.

The Automatic Flight Control System (AFCS) is used to guide the flight path of the aircraft by generating Flight Director (FD) commands that can be coupled to an optional three-axis Autopilot (AP), if S.O.O. 6221 is incorporated. The optional three-axis autopilot controls the aileron, elevator, and rudder servos to maintain the aircraft in a desired attitude. When equipped with the AP, the aircraft also has electric pitch trim and a yaw damper system.

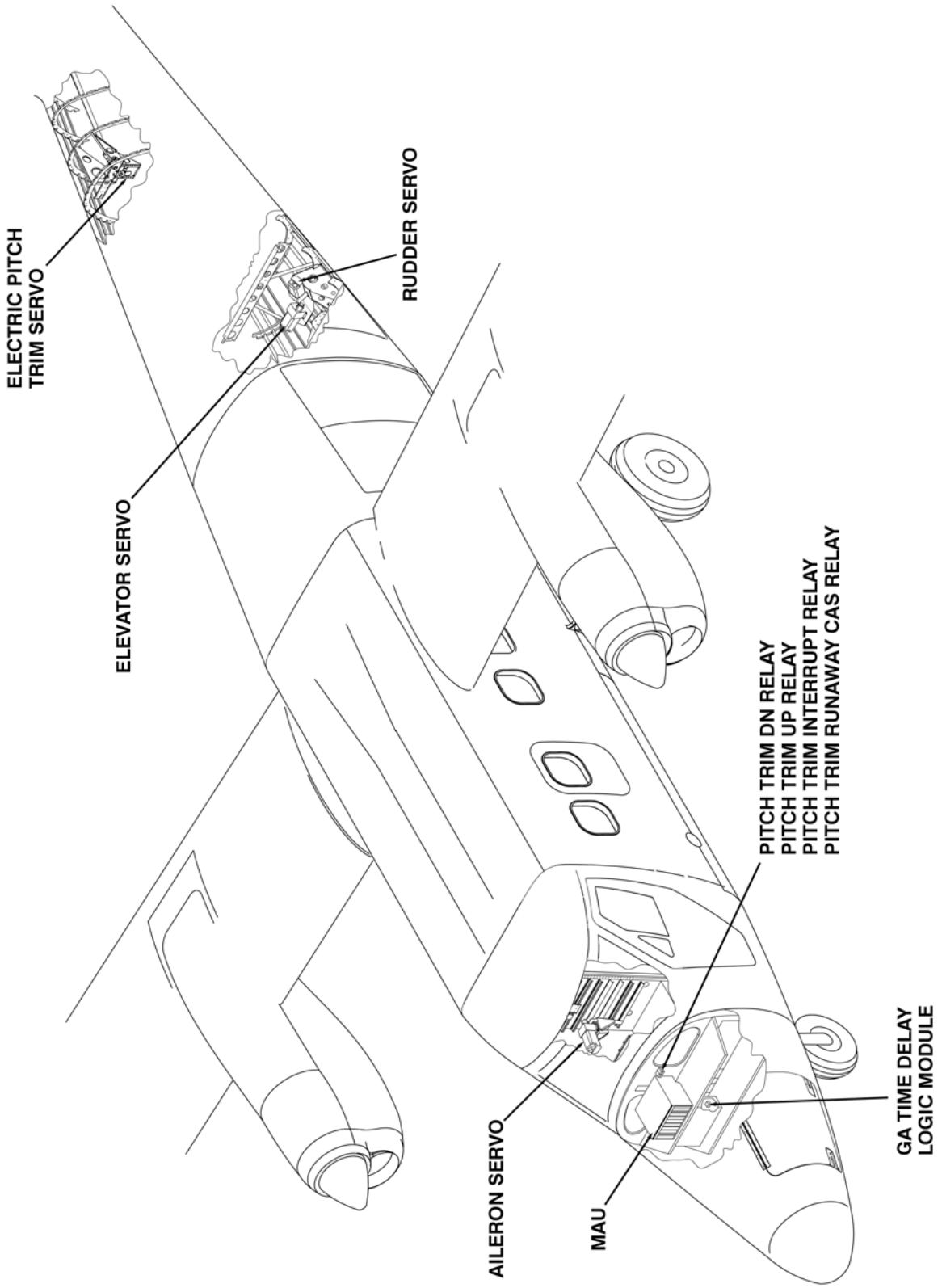
The AFCS consists of the following primary components:

- MAU AIOP Modules
- Flight Controller (FC)
- Control Wheel Switches
- Electric Pitch Trim Interrupt Relay
- Autopilot Servos (S.O.O. 6221)
- Electric Pitch Trim (S.O.O. 6221)
- Electric Pitch Trim Disable Switch (S.O.O. 6221)
- Electric Pitch Trim Up/Down Relays (S.O.O. 6221)
- GA Time Delay Logic Module (S.O.O. 6221)
- Pitch Trim Runaway CAS Relay (S.O.O. 6221)

The AP only receives vertical and lateral steering commands from the FD. The FD commands are used to generate required control deflections that are actuated by aileron, elevator and rudder servos, and automatic actuation of the electric pitch trim system. The servos are connected to the MAU via dual Controller Area Network (CAN) data buses. Commands received by the servo are processed within the servo to check for agreement prior to servo action. Both servo channels must agree in their monitoring of motor current, clutch solenoid engagement and motor position. The aileron, elevator and rudder servos incorporate internal electric clutch mechanisms that release when commanded by the MAU or when electrical power is removed (CB open or power failure). The pitch trim servo incorporates a slip-clutch mechanism which requires scheduled maintenance.

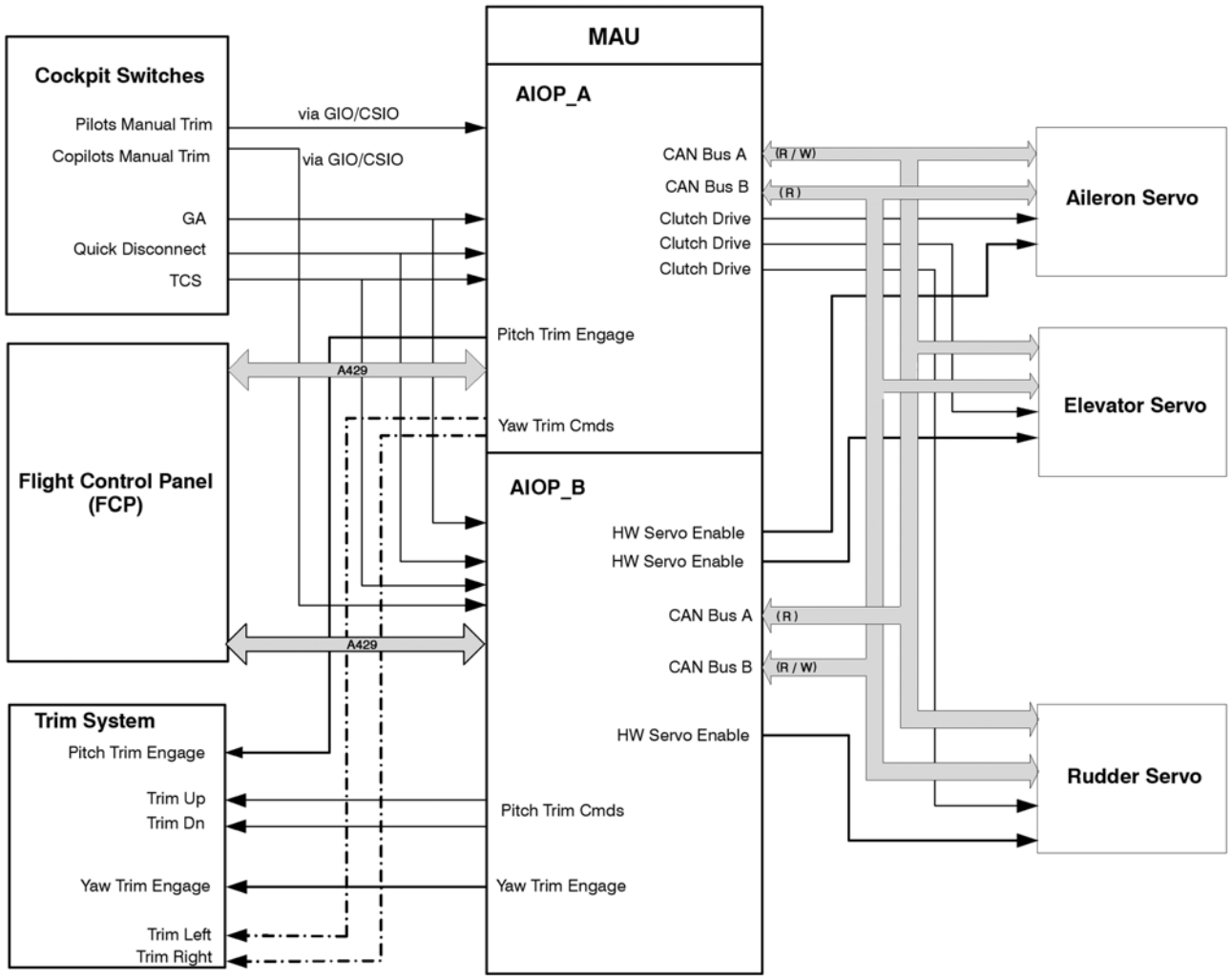
The AP controls the primary servos and trim servo through the MAU as shown in Figure 2. The FD generates AP commands utilizing data from the dual-channel ADAHRS, two GPS receivers, the Flight Management System (FMS), two MMDRs, a single Flight Controller (FC), engine data provided to Apex through two Data Acquisition Units (DAUs) and the aircraft's stall warning system. Figure 3 details a functional block diagram of the Apex AFCS.

The AFCS architecture operates around two parallel channels (A and B) within the Apex Modular Avionics Unit (MAU), both channels are required to be functional for normal AFCS operation. An AP disengage occurs if a miscompare is detected between AP channel commands.

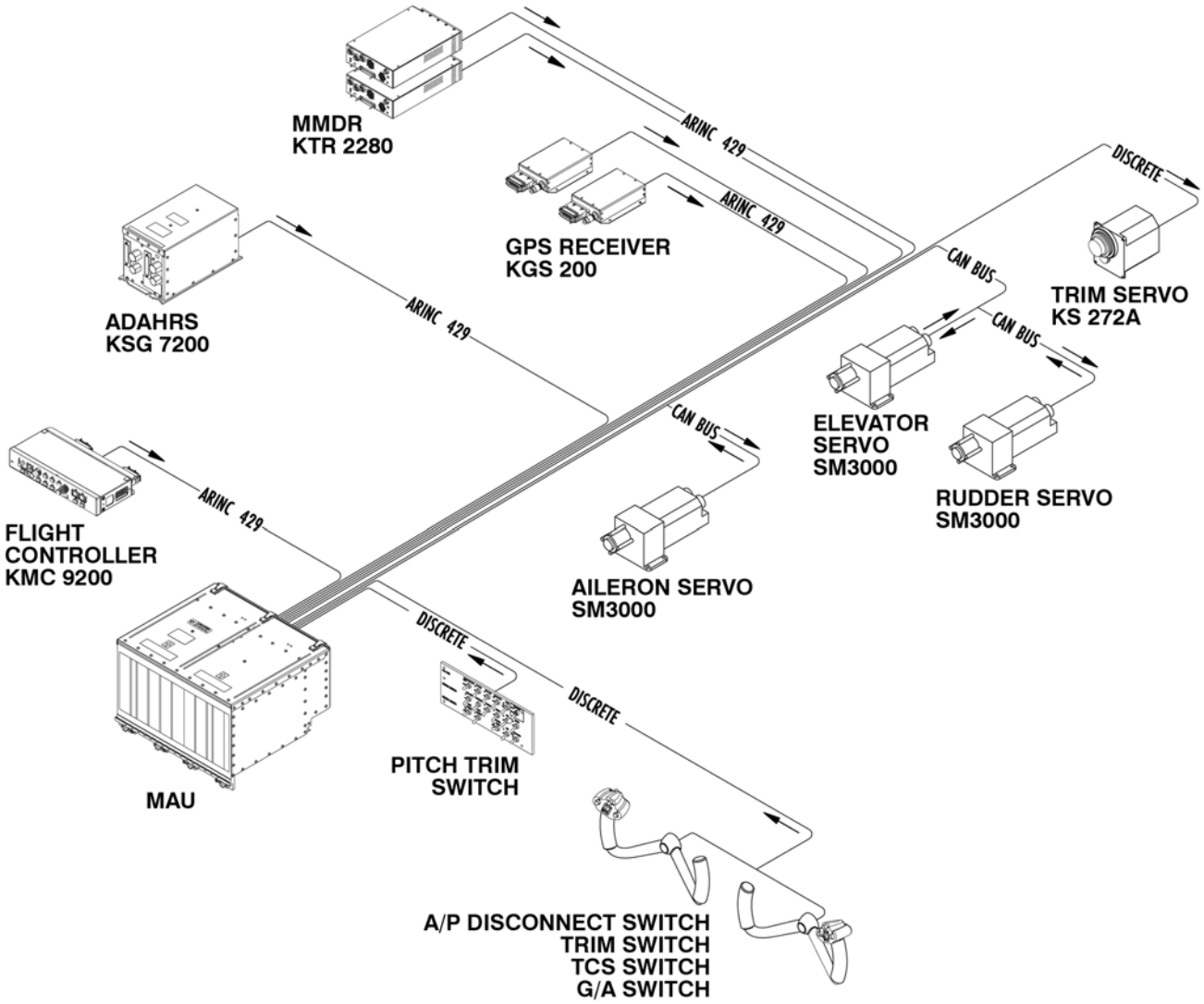


Location of Components
Figure 1

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Apex AFCS Architecture
Figure 2



Apex Automatic Flight Control System – Block Diagram
Figure 3

2. Description

A. Flight Controller (FC) (KMC 9200)

Refer to Figure 4 for details.

The KMC 9200 Flight Controller (FC) is used to select and activate various functions, including mode selection of the Flight Director (FD). Incorporating S.O.O. 6221 will add Autopilot (AP) coupled to the FD steering commands. The FC provides selection for all AFCS functions except for the following functions:

- Go Around (GA) mode
- Touch Control Steering (TCS)
- Quick Disconnect (AP DISC)

The functions via the Flight Controller include:

- L/R – Selects which PFD (Left or Right) is used for FD coupling. At power-up, the default setting is Left
- FD, AP, YD – Engage or disengage the Flight Director (FD) and with S.O.O. 6221 Autopilot (AP)/Yaw Damper (YD)
- Nose DN/Nose UP – Adjust target values in modes: PIT, SPD, VS
- HDG/T, NAV, BL, VS, SPD, ALT, VNAV, APR – Engage or disengage each associated lateral and vertical mode of the FD

Refer to 31-13-00, Apex Controllers – General data for Flight Controller (FC) details.



KMC 9200 Flight Controller
Figure 4

B. Flight Control Primary Servos (SM3000)

The aileron (roll), elevator (pitch) and rudder (yaw) fail passive, primary servos, move the associated aircraft control surfaces in response to signals from the AIOP (Actuator Input Output Processor) located within the MAU. Communication between the MAU and primary servos is over bidirectional CAN Buses, motor enable discrete and clutch enable discrete.

Refer to 22-10-11, Primary Servos (S.O.O. 6221) – Maintenance Practices for details on Primary Servos.

Refer to 22-10-12, Primary Servo Capstan (S.O.O. 6221) – Maintenance Practices for details on Capstan.

C. Pitch Trim Servo (KS 272A)

The pitch trim servo controls the elevator trim tab in response to signals from the AIOP within the MAU when autopilot is enabled, or directly from the Pitch Trim Up / Down Relays when the servo is manually operated with the control wheel pitch trim switches. (Refer to 22-20-00, Pitch Trim System – General Data).

D. Autopilot and Flight Director Switches

Refer to Figure 5 for details.

Four AFCS control wheel switches are located on the outboard side of the Pilot and Co-pilot control wheels. To accommodate AFCS switches, the radio XMIT switch has been relocated to the forward face of the hand wheel switch housing utilizing a different switch type from the legacy design.

1) Autopilot Disconnect Switch (AP only)

The Autopilot Disconnect switch marked AP DISC (red) (installed with AP) is located top-centre on the hand wheel switch cluster. AP DISC is a momentary switch which disconnects autopilot, yaw damper and the trim servo when pressed. Autopilot, yaw damper will remain disconnected after the AP DISC switch is released.

2) Go-Around Switch

The Go-around switch marked G/A (green) is located on the outboard side of the hand wheel switch cluster. G/A is a momentary switch which commands the FD into a wings level pitch up attitude and disengages the autopilot.

3) Touch Control Steering Switch

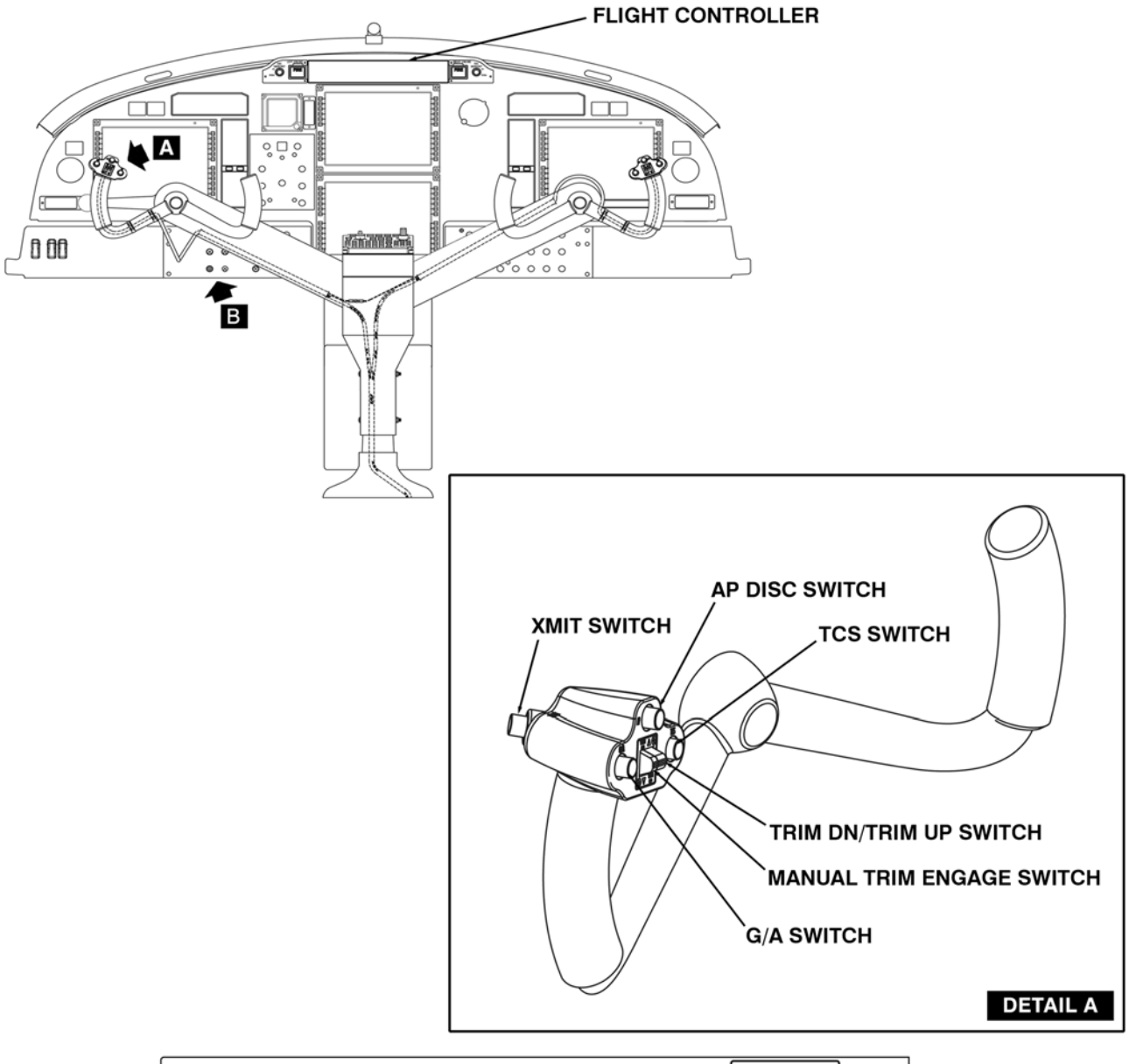
Touch control steering switch marked TCS (black) is located on the inboard side of the hand wheel switch cluster. TCS temporarily disconnects flight control servos and trim servo from the control surfaces when pressed. The system will maintain the aircraft flight condition (attitude, heading) when switch is released.

4) Trim Switches (AP only)

Trim switches marked TRIM DN and TRIM UP (installed with AP) is located in the centre of the hand wheel switch cluster. The trim switch manually controls the elevator trim servo. The switch has split controls, trim engage and trim direction. Both controls must be engaged to actuate the trim servo. Activation of the trim engage switch, while autopilot is engaged, will disengage autopilot.

5) Pitch Trim Switch (AP only)

A pitch trim disable switch marked PITCH TRIM is located on the pilot's switch panel. The switch has two positions, DISABLE and NORMAL. The switch must be in the NORMAL position (down) for the autopilot to function or for manual electronic control of the elevator trim servo. Selecting DISABLE will disconnect autopilot as well as disabling the manual electric pitch trim.



Autopilot Controls
Figure 5

3. Adjustment/Test

A. Flight Director (FD) Operational Test (Mod 6/2303)

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Wait several minutes for the aircraft electrical and avionic system to initialize.

3) Select the Flight Controller FD pushbutton, ensure that the LED above the button is ON.

4) On the Pilot and Co-pilot PFD, ensure the Flight Director bars are displayed and agree with the aircraft attitude.

5) On the Pilot and Co-pilot PFD, ensure that a green ROL and green PIT are displayed as active modes.

6) On the Flight Controller, press and hold the NOSE UP pushbutton, ensure the Flight Director bars move up.

7) On the Flight Controller, press and hold the NOSE DN pushbutton, ensure the Flight Director bars move down.

8) On the Flight Controller, press the HEADING rotary control to centre the heading bug.

9) On the Flight controller, press the HDG/T pushbutton, ensure that the LED above the button is ON.

10) On the Pilot and Co-pilot PFD, ensure the PFD heading bug is displayed in magenta.

11) Use the Flight Controller heading rotary knob to move the heading bug 20 degrees left of the aircraft heading, ensure the Flight Director bars follow the heading bug to the left.

12) Use the Flight Controller heading rotary knob to move the heading bug 20 degrees right of the aircraft heading, ensure the Flight Director bars follow the heading bug to the right.

13) On the Flight Controller, press the HDG/T pushbutton, ensure that the LED above the button is OFF.

14) Select the Pilot Go-Around (G/A) switch, ensure green GA indication on the Pilot and Co-Pilot PFD and the Flight Director bars on the Pilot and Co-pilot PFD move up slightly above the current aircraft attitude.

15) Select the Pilot Touch Control Steering (TCS) switch, ensure an amber TCS and green PIT indication on the Pilot and Co-pilot PFD and FD bars return to centre.

16) Select the Co-pilot Go-Around (G/A) switch, ensure green GA indication on the Pilot and Co-Pilot PFD and the Flight Director bars on the Pilot and Co-pilot PFD move up slightly above the current aircraft attitude.

17) Select the Co-pilot Touch Control Steering (TCS) switch, ensure an amber TCS and green PIT indication on the Pilot and Co-pilot PFD and FD bars return to centre.

18) Select the Flight Controller FD pushbutton, ensure that the LED above the button is OFF.

19) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Automatic Flight Control System (AFCS) Operational (Routine) Test (S.O.O. 6221)

1) Ensure that AP SERVOS (P1) (Left Bus) and PITCH TRIM (N1) (Left Bus) circuit breakers on the avionics circuit breaker panel are closed.

2) Ensure the PITCH TRIM switch is in the NORMAL position.

3) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

4) Wait several minutes for the aircraft electrical and avionics systems to initialize. During initialization, a Power-on Built In Test (PBIT) test will be completed on the Pitch Trim servo system.

5) Confirm AP 1 Fail and YD 1 Fail CAS messages extinguish and are not displayed after the avionics systems initialize.

6) Select the Pilot's control wheel switch to the TRIM UP position, confirm the trim wheel travels TRIM UP.

7) Select the Pilot's control wheel switch to the TRIM DN position, confirm the trim wheel travels TRIM DN.

8) Select the Co-pilot's control wheel switch to the TRIM UP position, confirm the trim wheel travels TRIM UP.

9) Select the Co-pilot's control wheel switch to the TRIM DN position, confirm the trim wheel travels TRIM DN.

10) Select and hold the Pilot's control wheel switch to either the TRIM UP or TRIM DN position and press and hold the AP DISC simultaneously, confirm the control wheel stops. Release the AP DISC, confirm the trim wheel travels resumes. Release the trim switch.

11) Select and hold the Co-pilot's control wheel switch to either the TRIM UP or TRIM DN position and press and hold the AP DISC simultaneously, confirm the control wheel stops. Release the AP DISC, confirm the trim wheel travels resumes. Release the trim switch.

12) Select and hold the Pilot's control wheel trim switch to either the TRIM UP or TRIM DN position and select the PITCH TRIM switch to the DISABLE position, confirm the control wheel stops. Release the trim switch. Confirm the AP 1 Fail CAS message illuminates.

13) Select the PITCH TRIM switch to the NORMAL position, confirm the AP 1 Fail CAS message extinguishes.

14) Select the Flight Controller YD pushbutton, ensure the following:

    a) LED above the Flight Controller YD pushbutton is ON.

    b) On the Pilot and Co-pilot PFD, a green YD is displayed.

    c) The yaw servo is engaged by confirming the rudder pedals are firm.

        1 On the Pilot control wheel select the AP DISC. Ensure the following:

            1 LED above the Flight Controller YD pushbutton is OFF.

            2 On the Pilot and Co-pilot PFD, a green YD is not displayed.

            3 The yaw servo is disengaged by confirming the rudder pedals can be moved freely.

15) Select the Flight Controller YD pushbutton, ensure the following:

    a) LED above the Flight Controller YD pushbutton is ON.

    b) On the Pilot and Co-pilot PFD, a green YD is displayed.

    c) The yaw servo is engaged by confirming the rudder pedals are firm.

        1 On the Co-pilot control wheel select the AP DISC. Ensure the following:

            1 LED above the Flight Controller YD pushbutton is OFF.

            2 On the Pilot and Co-pilot PFD, a green YD is not displayed.

            3 The yaw servo is disengaged by confirming the rudder pedals can be moved freely.

16) Select the Flight Controller YD pushbutton, ensure the following:

a) LED above the Flight Controller YD pushbutton is ON.

b) On the Pilot and Co-pilot PFD, a green YD is displayed.

c) The yaw servo is engaged by confirming the rudder pedals are firm.

1 On the Pilot control wheel select the GA button. Ensure the following:

1 LED above the Flight Controller YD pushbutton is OFF.

2 On the Pilot and Co-pilot PFD, green YD is not displayed.

3 The yaw servo is disengaged by confirming the rudder pedals can be moved freely.

4 On the Pilot and Co-pilot PFD, a green ROL and GA is displayed.

5 On the Pilot and Co-pilot PFD, the flight director bars move up slightly above the current aircraft attitude.

6 Press the Pilot TCS to cancel the GA mode.

17) Select the Flight Controller YD pushbutton, ensure the following:

a) LED above the Flight Controller YD pushbutton is ON.

b) On the Pilot and Co-pilot PFD, a green YD is displayed.

c) The yaw servo is engaged by confirming the rudder pedals are firm.

1 On the Co-pilot control wheel select the GA button. Ensure the following:

1 LED above the Flight Controller YD pushbutton is OFF.

2 On the Pilot's and Co-pilot's PFD, green YD is not displayed.

3 The yaw servo is disengaged by confirming the rudder pedals can be moved freely.

4 On the Pilot's and Co-pilot's PFD, a green ROL and GA is displayed.

5 On the Pilot's and Co-pilot's PFD, the flight director bars move up slightly above the current aircraft attitude.

6 Press the Co-pilot TCS to cancel the GA mode.

18) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

C. AFCS Function Test (S.O.O. 6221)

WARNING **WHEN OPERATING THE AUTOPILOT, ENSURE THE FLIGHT CONTROLS ARE CLEAR AND ALL RIGGING PINS ARE REMOVED.**

Note If any power interruption has occurred prior to completion of the MAU's internal self-test, AP 1 Fail CAS message may persist longer than 2 minutes. If this occurs, an OFF/ON cycle of the aircraft electrical power is required to allow the MAU to reboot without power interruptions.

Note When the Autopilot is engaged, ensure it does not drive the servos against the stops. Autopilot is to be engaged for short periods of time for all ground tests. If at any time the Autopilot begins to drive a servo against the stop, disengage Autopilot and discontinue the test. After the flight controls, rudder pedal, and trims are reset to the center position, the Autopilot can be reengaged and the test steps continued.

1) AFCS Function Test

a) Ensure that AP SERVOS (P1) (Left Bus) and PITCH TRIM (N1) (Left Bus) circuit breakers, on the avionics circuit breaker panel are closed.

b) Ensure the PITCH TRIM switch is in the NORMAL position.

c) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

d) Wait several minutes for the aircraft electrical and avionics systems to initialize. During initialization, a Power-on Built In Test (PBIT) test will be completed on the Pitch Trim servo system.

e) Confirm AP 1 Fail and YD 1 Fail CAS messages extinguish and are not displayed after the avionics systems initialize.

f) Attach a pitot/static test set (refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data) and increase airspeed to >60 knots to enable autopilot functionality.

g) Ensure the flight controls move freely in roll, pitch and yaw. Set the flight controls and rudder pedals to the center position.

h) On the Flight Controller select the L/R pushbutton. Ensure L/R can be selected to either the Pilot position (left) or Co-pilot position (right). Upon completion, select the Pilot (L) position.

i) Select the Flight Controller FD pushbutton, ensure that the LED above the button is ON.

j) On the Pilot and Co-pilot PFD, ensure the flight director bars are displayed and agree with the aircraft attitude.

k) On the Pilot and Co-pilot PFD, ensure that a green ROL and green PIT are displayed as active modes.

l) On the Flight Controller, press the HEADING rotary control to center the heading bug and press the HDG/T pushbutton. Ensure that the LED above the HDG/T pushbutton is ON.

m) On the Pilot and Co-pilot PFD, ensure the PFD heading bug is displayed in magenta.

n) Select the Flight Controller AP pushbutton, ensure that the LED above the pushbutton is ON and that the LED above the YD pushbutton is ON.

o) On the Pilot and Co-pilot PFD, ensure green AP and YD are displayed.

p) Ensure that the autopilot servos engage and the flight controls become firm.

q) Use the Flight Controller heading rotary knob to move the heading 5 degrees left of the aircraft heading, ensure the Flight Director bars and flight controls follow the heading bug to the left.

r) Use the Flight Controller heading rotary knob to move the heading 5 degrees right of the aircraft heading, ensure the Flight Director bars and flight controls follow the heading bug to the right.

s) On the Flight Controller, press the HDG/T pushbutton, ensure that the LED above the button is OFF.

t) On the Flight Controller, repeatedly select the NOSE UP pushbutton, ensure the Flight Director bars move up and flight controls move aft.

u) On the Flight Controller, repeatedly select the NOSE DN pushbutton, ensure the Flight Director bars move down and flight controls move forward.

v) On the FC select the AP pushbutton, ensure the LED above the AP is OFF. Ensure the LED above the YD is ON.

w) On the Pilot and Co-pilot PFD, ensure AP flashes red and red inverse and the aural warning cavalry charge persists.

x) Ensure that the autopilot roll and pitch servos disengage and the flight controls move freely in roll and pitch.

y) On the Pilot or Co-pilot control wheel select the AP DISC, ensure that the red flashing AP clears, and the cavalry charge aural mutes.

z) On the FC, ensure the green LED above the YD button is OFF and the YAW servo has disengaged by moving the rudder pedals freely.

aa) Reduce the airspeed on the Pitot/Static test set to bring the aircraft into GND mode.

ab) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) AFCS Engage/Disengage Test

a) Ensure that AP SERVOS (P1) (Left Bus) and PITCH TRIM (N1) (Left Bus) circuit breakers on the avionics circuit breaker panel are closed.

b) Ensure the PITCH TRIM switch is in the NORMAL position.

c) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

d) Wait several minutes for the aircraft electrical and avionics systems to initialize. During initialization, a PBIT test will be completed on the Pitch Trim servo system.

e) Confirm AP 1 Fail and YD 1 Fail CAS messages are not displayed after the avionics systems initialize.

f) Attach a pitot/static test set (refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data) and increase airspeed to >60 knots to enable autopilot functionality.

g) Set the flight controls and rudder pedals to the center position. On the FC, select the AP pushbutton. Ensure the following:

1 The LED above the Flight Controller AP pushbutton is ON.

2 The LED above the Flight Controller YD pushbutton is ON.

3 The Pilot and Co-pilot PFD green AP and YD are displayed.

4 The autopilot servo clutches engage and the flight controls are firm.

h) On the Pilot control yoke, press and release the Quick Disconnect (AP DISC). Ensure the following:

1 The cavalry charge aural warning is continually heard.

2 The LED above the Flight Controller AP pushbutton is OFF.

3 The LED above the Flight Controller YD pushbutton is OFF.

4 The Pilot and Co-pilot PFD AP flashes red and red inverse.

5 On the Pilot control yoke, press and release the Quick Disconnect (AP DISC). Ensure on the Pilot and Co-pilot PFDs the flashing red AP and green YD are not displayed. Ensure the cavalry charge is muted.

6 The autopilot servo clutches disengage and the flight controls can be moved freely in roll, pitch and yaw. Set the flight controls and rudder pedals to the center position.

i) On the FC, select the AP pushbutton. Ensure the following:

1 The LED above the Flight Controller AP pushbutton is ON.

2 The LED above the Flight Controller YD pushbutton is ON.

3 The Pilot and Co-pilot PFD green AP and YD are displayed.

4 The autopilot servo clutches engage and the flight controls are firm.

j) On the Co-pilot control yoke, press and release the AP DISC. Ensure the following:

1 The cavalry charge aural warning is continually heard.

2 The LED above the Flight Controller AP pushbutton is OFF.

3 The LED above the Flight Controller YD pushbutton is OFF.

4 The Pilot and Co-pilot PFD AP flashes red and red inverse.

5 On the Co-pilot control yoke, press and release the Quick Disconnect (AP DISC). Ensure on the Pilot and Co-pilot PFDs the flashing red AP and green YD are not displayed. Ensure the cavalry charge is muted.

6 The autopilot servo clutches disengage and the flight controls can be moved freely in roll, pitch and yaw. Set the flight controls and rudder pedals to the center position.

k) On the FC, select the AP pushbutton. Ensure the following:

1 The LED above the Flight Controller AP pushbutton is ON.

2 The LED above the Flight Controller YD pushbutton is ON.

3 The Pilot and Co-pilot PFD green AP and YD are displayed.

4 The autopilot servo clutches engage and the flight controls are firm.

l) On the Pilot control yoke, press, briefly hold, and release the pitch trim switch to either the TRIM DN or TRIM UP position. Ensure the following:

1 The cavalry charge aural warning is continually heard.

2 The LED above the Flight Controller AP pushbutton is OFF.

3 The LED above the Flight Controller YD pushbutton is ON.

4 The Pilot and Co-pilot PFD AP flashes red and red inverse.

5 On the Pilot control yoke, press and release the Quick Disconnect (AP DISC). Ensure on the Pilot and Co-pilot PFDs the flashing red AP and green YD are not displayed. Ensure the cavalry charge is muted. Ensure the LED above the Flight Controller YD pushbutton is OFF.

6 The autopilot servo clutches disengage and the flight controls can be moved freely in roll, pitch and yaw. Set the flight controls and rudder pedals to the center position.

m) On the FC, select the AP pushbutton. Ensure the following:

1 The LED above the Flight Controller AP pushbutton is ON.

2 The LED above the Flight Controller YD pushbutton is ON.

3 The Pilot and Co-pilot PFD green AP and YD are displayed.

4 The autopilot servo clutches engage and the flight controls are firm.

n) On the Co-pilot control yoke, press, briefly hold, and release the TRIM DN or TRIM UP position. Ensure the following:

1 The cavalry charge aural warning is continually heard.

2 The LED above the Flight Controller AP pushbutton is OFF.

3 The LED above the Flight Controller YD pushbutton is ON.

4 The Pilot and Co-pilot PFD AP flashes red and red inverse.

5 On the Co-pilot control yoke, press and release the Quick Disconnect (AP DISC). Ensure on the Pilot and Co-pilot PFDs the flashing red AP and green YD are not displayed. Ensure the cavalry charge is muted. Ensure the LED above the Flight Controller YD pushbutton is OFF.

6 The autopilot servo clutches disengage and the flight controls can be moved freely in roll, pitch and yaw. Set the flight controls and rudder pedals to the centre position.

o) On the FC, select the AP pushbutton. Ensure the following:

1 The LED above the Flight Controller AP pushbutton is ON.

2 The LED above the Flight Controller YD pushbutton is ON.

3 The Pilot and Co-pilot PFD green AP and YD are displayed.

4 The autopilot servo clutches engage and the flight controls are firm.

p) On the Pilot's switch panel, select the PITCH TRIM switch to the DISABLE (up) position. Ensure the following:

1 The cavalry charge aural warning is continually heard.

2 The LED above the Flight Controller AP pushbutton is OFF.

3 The LED above the Flight Controller YD pushbutton is ON.

4 The Pilot and Co-pilot PFD AP flashes red and red inverse.

5 On the Pilot or Co-pilot control yoke, press and release the Quick Disconnect (AP DISC). Ensure on the Pilot and Co-pilot PFDs the flashing red AP and green YD are not displayed. Ensure the cavalry charge is muted. Ensure the LED above the Flight Controller YD pushbutton is OFF.

6 The autopilot servo clutches disengage and the flight controls can be moved freely in roll, pitch and yaw.

7 Confirm AP 1 Fail message is displayed.

q) On the Pilot's switch panel, select the PITCH TRIM switch to the NORMAL (down) position. Ensure the following:

1 Confirm AP 1 Fail messages is not displayed.

r) Set the flight controls and rudder pedals to the center position. On the FC, select the AP pushbutton. Ensure the following:

1 The LED above the Flight Controller AP pushbutton is ON.

2 The LED above the Flight Controller YD pushbutton is ON.

3 The Pilot and Co-pilot PFD green AP and YD are displayed.

s) On the avionics circuit breaker panel, open the PITCH TRIM (N1) circuit breaker. Ensure the following:

1 The cavalry charge aural warning is continually heard.

2 The LED above the Flight Controller AP pushbutton is OFF.

3 The LED above the Flight Controller YD pushbutton is ON.

4 The Pilot and Co-pilot PFD AP flashes red and red inverse.

5 On the Pilot or Co-pilot control yoke, press and release the Quick Disconnect (AP DISC). Ensure on the Pilot and Co-pilot PFDs the flashing red AP and green YD are not displayed. Ensure the cavalry charge is muted. Ensure the LED above the Flight Controller YD pushbutton is OFF.

6 The autopilot servo clutches disengage and the flight controls can be moved freely in roll, pitch and yaw. Set the flight controls and rudder pedals to the centre position.

7 Confirm AP 1 Fail message is displayed.

t) On the avionics circuit breaker panel, close the PITCH TRIM (N1) circuit breaker. Ensure the following:

1 Confirm AP 1 Fail message is not displayed.

u) Set the flight controls and rudder pedals to the center position. On the FC, select the AP pushbutton. Ensure the following:

1 The LED above the Flight Controller AP pusbutton is ON.

2 The LED above the Flight Controller YD pusbutton is ON.

3 The Pilot and Co-pilot PFD green AP and YD are displayed.

v) On the avionics circuit breaker panel, open the AP SERVOS (P1) circuit breaker. Ensure the following:

1 The cavalry charge aural warning is continually heard.

2 The LED above the Flight Controller AP pushbutton is OFF.

3 The LED above the Flight Controller YD pushbutton is OFF.

4 The Pilot and Co-pilot PFD AP flashes red and red inverse. Ensure amber YD is displayed.

5 On the Pilot or Co-pilot control yoke, press and release the Quick Disconnect (AP DISC). Ensure on the Pilot and Co-pilot PFDs the flashing red AP is not displayed, and amber YD is displayed. Ensure the cavalry charge is muted.

6 The autopilot servo clutches disengage and the flight controls can be moved freely in roll, pitch and yaw.

7 Confirm AP 1 Fail and YD 1 Fail messages are displayed.

w) On the avionics circuit breaker panel, close the AP SERVOS (P1) circuit breaker. Ensure the following:

1 Confirm AP 1 Fail and YD 1 Fail messages are not displayed.

2 The Pilot and Co-pilot PFD, ensure the amber YD is not displayed.

x) Reduce the airspeed on the Pitot/Static test set to bring the aircraft into GND mode.

y) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

3) AFCS Touch Control Steering (TCS) Test

a) Ensure that AP SERVOS (P1) (Left Bus) and PITCH TRIM (N1) (Left Bus) circuit breakers, on the avionics circuit breaker panel are closed.

b) Ensure the PITCH TRIM switch is in the NORMAL position.

c) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

d) Wait several minutes for the aircraft electrical and avionics systems to initialize. During initialization, a PBIT test will be completed on the Pitch Trim servo system.

e) Confirm AP 1 Fail and YD 1 Fail CAS messages extinguish and are not displayed after the avionics systems initialize.

f) Attach a pitot/static test set (refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data) and increase airspeed to >60 knots to enable autopilot functionality.

g) Set the flight controls and rudder pedals to the center position. On the FC, select the AP pushbutton. Ensure the following:

1 The LED above the Flight Controller AP pushbutton is ON.

2 The LED above the Flight Controller YD pushbutton is ON.

3 The Pilot and Co-pilot PFD green AP and YD are displayed.

4 The autopilot servo clutches engage and the flight controls are firm.

h) On the Pilot control yoke, press and hold the TCS pushbutton. Ensure the following:

1 On the Pilot and Co-pilot PFD's the AP caption changes from green AP to an amber TCS.

2 The pitch and roll servos disengage and the flight controls can be moved freely. Set the flight controls to the center position.

i) Release the Pilot TCS pushbutton. Ensure the following:

1 On the Pilot and Co-pilot PFD's the amber TCS caption changes to a green AP.

2 The pitch and roll servos re-engage and the flight controls are firm.

j) On the Co-pilot control yoke, press and hold the TCS pushbutton. Ensure the following:

1 On the Pilot and Co-pilot PFD's the AP caption changes from green AP to an amber TCS.

2 The pitch and roll servos disengage and the flight controls can be moved freely. Set the flight controls to the center position.

k) Release the Co-pilot TCS pushbutton. Ensure the following:

1 On the Pilot and Co-pilot PFD's the amber TCS caption changes to a green AP.

2 The pitch and roll servos re-engage and the flight controls are firm.

l) Reduce the airspeed on the Pitot/Static test set to bring the aircraft into GND mode.

m) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

4) AFCS Trim Sense Test

a) Ensure that AP SERVOS (P1) (Left Bus) and PITCH TRIM (N1) (Left Bus) circuit breakers, on the avionics circuit breaker panel are closed.

b) Ensure the PITCH TRIM switch is in the NORMAL position.

c) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

d) Wait several minutes for the aircraft electrical and avionics systems to initialize. During initialization, a PBIT test will be completed on the Pitch Trim servo system.

e) Confirm AP 1 Fail and YD 1 Fail CAS messages are not displayed after the avionics systems initialize.

f) Attach a pitot/static test set (refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data) and increase airspeed to >60 knots to enable autopilot functionality.

g) Move the flight controls to approximately the center position.

h) On the FC, select the AP pushbutton. Ensure the following:

1 The LED above the Flight Controller AP pushbutton is ON.

2 The LED above the Flight Controller YD pushbutton is ON.

3 The Pilot and Co-pilot PFD green AP and YD are displayed.

4 The autopilot servo clutches engage and the flight controls are firm.

i) Hold the flight control yoke and gently turn it to the right with enough force to turn the yoke, but not enough force to override the servo.

j) Ensure after approximately 15 seconds, the AP Hold LH Wing Down caution CAS message displays.

k) Release the flight control yoke, ensure the AP Hold LH Wing Down caution CAS message extinguishes.

l) Hold the flight control yoke and gently turn it to the left with enough force to turn the yoke, but not enough force to override the servo.

m) Ensure after approximately 15 seconds, the AP Hold RH Wing Down caution CAS message displays.

n) Release the flight control yoke, ensure the AP Hold RH Wing Down caution CAS message extinguishes.

o) Hold the flight controls and push them forward with enough force to move the controls, but not enough force to override the servo. Ensure the following:

1 The pitch trim running indicator (green dot) shows on the MFD.

2 The pitch trim indicator drives nose UP.

3 The manual pitch trim wheel drives nose UP.

4 After approximately 10 seconds the AP Hold Nose Up caution CAS message is displayed.

p) Release the flight controls. Ensure the following:

1 The pitch trim running indicator (green dot) extinguishes on the MFD.

2 The pitch trim indicator stops moving.

3 The manual pitch trim wheel stops moving.

4 The AP Hold Nose Up caution CAS message extinguishes.

q) Hold the flight controls and pull them back with enough force to move the controls, but not enough to override the servo. Ensure the following:

1 The pitch trim running indicator (green dot) shows on the MFD.

2 The pitch trim indicator drives nose DOWN.

3 The manual pitch trim wheel drives nose DOWN.

4 After approximately 10 seconds the AP Hold Nose Down caution CAS message is displayed.

r) Release the flight controls. Ensure the following:

1 The pitch trim running indicator (green dot) extinguishes on the MFD.

2 The pitch trim indicator stops moving.

3 The manual pitch trim wheel stops moving.

4 The AP Hold Nose Down caution CAS message extinguishes.

s) On the FC, select AP to disconnect the autopilot and select the AP DISC on the control yoke to mute the cavalry charge aural alert.

t) Reduce the airspeed on the Pitot/Static test set to bring the aircraft into GND mode.

u) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

4. De-activation/Re-activation

A. De-activation of Autopilot (S.O.O. 6221)

Note Do not pull AP SERVOS circuit breaker (P1) on the Avionics CB Panel. This circuit breaker must be left engaged, as it is required for normal Flight Director operation.

Note Leave all electrical connectors installed on AP Servos and Pitch Trim Servo.

Note Leave bridle cables installed and tensioned on AP Servos and Pitch Trim Servo.

Note With Autopilot disabled, the following tests are not required:

- AFCS Operational (Routine) Test (S.O.O. 6221), refer to Para 3.B
- AFCS Function Test (Mod S.O.O. 6221), refer to Para 3.C
- KS 272A Pitch Trim Servo System, refer to 22-20-00, Pitch Trim System (S.O.O. 6221) – General Data, Para 3.A.

1) Ensure aircraft power is OFF.

2) Pull PITCH TRIM circuit breaker (N1) and secure with safety clip.

3) Select DISABLE position on PITCH TRIM switch (refer to Figure 6).

4) Install ELECTRIC PITCH TRIM INOP placard at the location shown in Figure 6, to the right of PITCH TRIM switch. Text height to be minimum 0.25” in height and clearly legible.

5) Install AUTOPILOT AND ELECTRIC PITCH TRIM DISABLED placard at location shown in Figure 6, on bezel of upper MFD screen, below Flight Controller. Text height to be minimum 0.40” in height and clearly legible.

6) Install Apex Option File with Autopilot Options disabled, as per Section 45-00-01, Apex Options and Settings File – Load Procedures.

7) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

8) Wait several minutes for the aircraft electrical and avionics systems to initialize.

9) Attach a Pitot/Static test set (refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data) and increase airspeed to above 60 knots to AIR mode to enable autopilot functionality.

10) Confirm the Flight Controller YD does not engage when selected:

a) LED above the Flight Controller YD pushbutton is OFF.

b) On the Pilot’s and Co-pilot’s PFD, a green YD is not displayed.

c) The yaw servo is disengaged by confirming the rudder pedals can be moved freely.

d) Confirm FUNCTION UNAVAILABLE Status CAS message.

11) Confirm Flight Controller AP pushbutton does not engage when selected:

a) LED above the Flight Controller AP pushbutton is OFF.

b) LED above the Flight Controller YD pushbutton is OFF.

c) Confirm FUNCTION UNAVAILABLE Status CAS message.

12) Reduce the airspeed on the Pitot/Static test set to bring the aircraft into GROUND mode.

13) Do operational test of the Flight Director, as per Para 3.A.

B. Re-activation of Autopilot (S.O.O. 6221)

1) Install Apex Option File with Autopilot Options enabled, as per Section 45-00-01, Apex Options and Settings File – Load Procedures.

2) Remove safety clips and close PITCH TRIM circuit breaker (N1).

3) Remove ELECTRIC PITCH TRIM INOP placard at location shown in Figure 6, to the right of PITCH TRIM switch.

4) Remove AUTOPILOT AND ELECTRIC PITCH TRIM DISABLED placard at location shown in Figure 6, on bezel of upper MFD screen, below Flight Controller.

5) Select NORMAL position on PITCH TRIM switch (refer to Figure 6).

6) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

7) Wait several minutes for the aircraft electrical and avionics systems to initialize.

8) Attach a Pitot/Static test set (refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data) and increase airspeed to above 60 knots to AIR mode to enable autopilot functionality.

9) Select the Flight Controller YD, ensure the following:

a) LED above the Flight Controller YD pushbutton is ON.

b) On the Pilot's and Co-pilot's PFD, a green YD is displayed.

c) The yaw servo is engaged by confirming the rudder pedals are firm.

10) Confirm Flight Controller AP pushbutton does engage when selected:

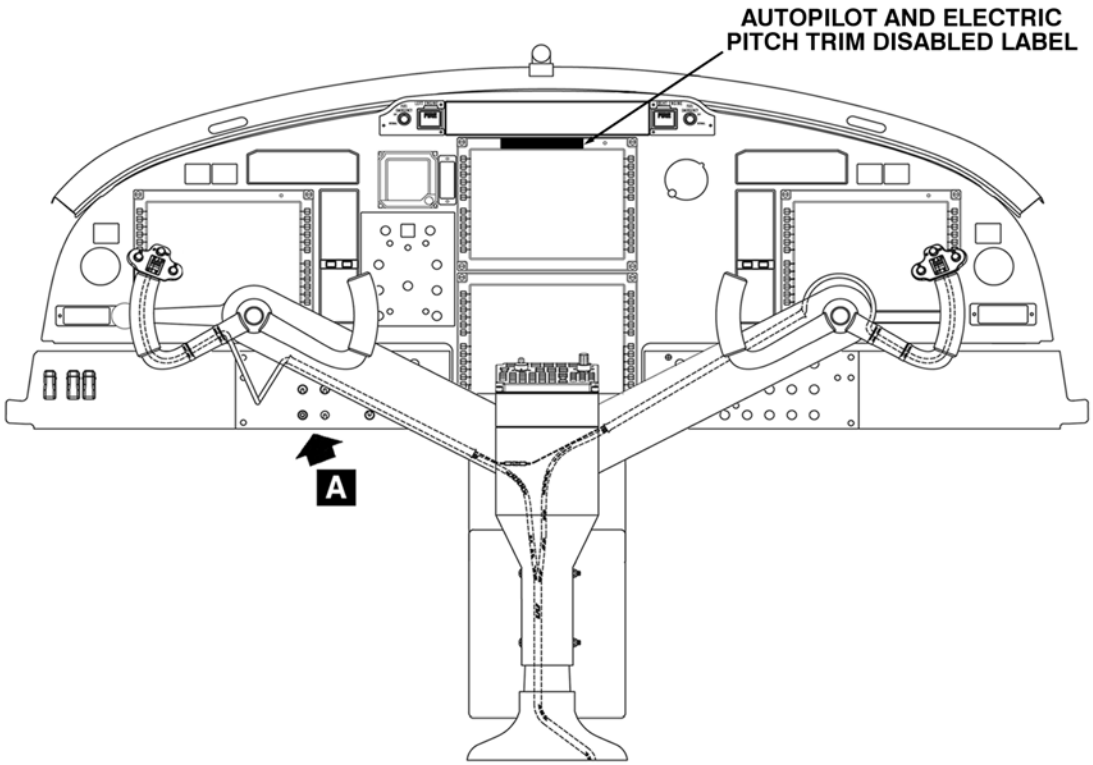
a) LED above the Flight Controller AP pushbutton is ON.

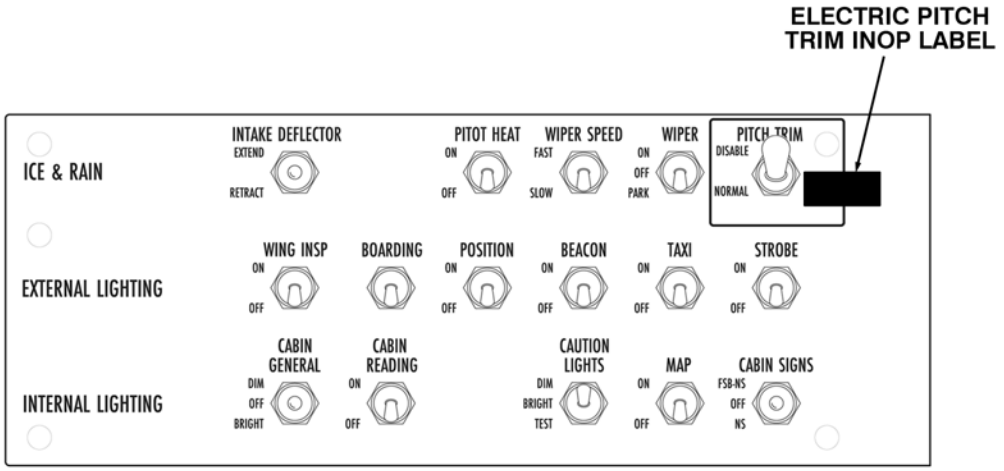
b) LED above the Flight Controller YD pushbutton is ON.

c) The yaw servo is engaged by confirming the rudder pedals are firm.

11) Do a operational test of the Flight Director, as per Para 3.A.

12) Do a function test of the AFCS, as per Para 3.C.



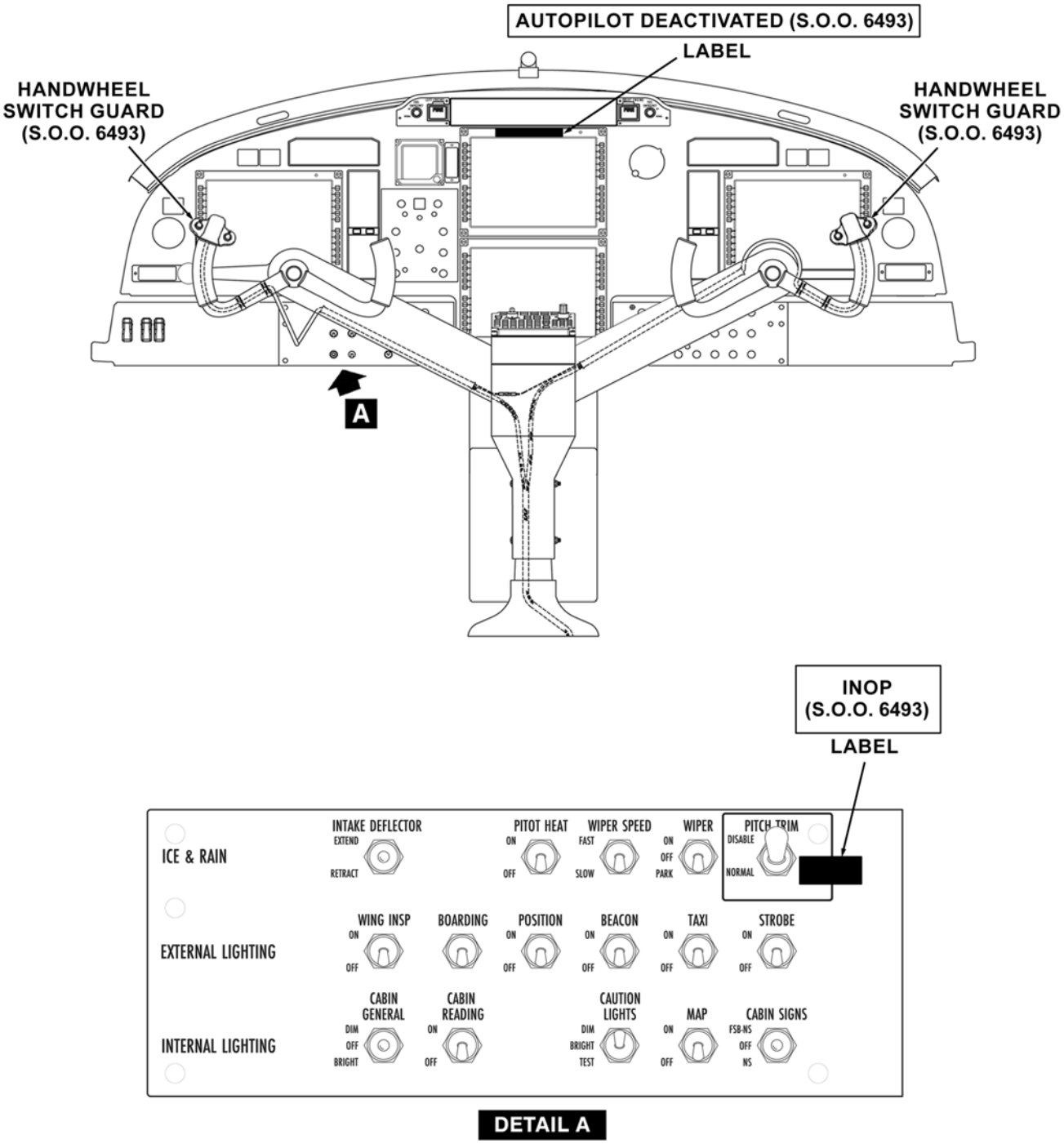


DETAIL A

Installation of ELECTRIC PITCH TRIM INOP and AUTOPILOT AND ELECTRIC PITCH TRIM DISABLED Placards
Figure 6

C. Deactivate S.O.O. 6221 Autopilot (S.O.O. 6493)

S.O.O. 6493 deactivates the autopilot as installed by S.O.O. 6221. This modification removes the PITCH TRIM (N1) circuit breaker and caps and stows the associated circuit breaker wiring. A Honeywell Options file with Autopilot not enabled is installed and a switch guard is installed on the pilot and co-pilot handwheels. DEACTIVATE AUTOPILOT (S.O.O. 6493) and INOP (S.O.O. 6493) placards are added on the instrument panel. Refer to Figure 7.



Installation of S.O.O 6493 Placards and Handwheel Switch Guards
Figure 7

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